

Chemistry Chapter 17 Practice Quiz 5 — Comprehensive Review

Multiple Choice

Identify the choice that best completes the statement or answers the question.

- ___ 1. At chemical equilibrium,
- | | |
|---|---|
| a. the forward reaction rate is greater than the reverse rate | b. the concentrations of all substances are equal |
| c. the forward and reverse reaction rates are equal and concentrations are stable | d. all reactants have been converted to products |
- ___ 2. For $\text{H}_2\text{O}(\text{g}) + \text{C}(\text{s}) \rightleftharpoons \text{H}_2(\text{g}) + \text{CO}(\text{g})$, which is the correct K_{eq} ?
- | | |
|---|---|
| a. $K_{\text{eq}} = [\text{H}_2\text{O}][\text{C}] / [\text{H}_2][\text{CO}]$ | b. $K_{\text{eq}} = [\text{H}_2][\text{CO}] / [\text{H}_2\text{O}]$ |
| c. $K_{\text{eq}} = [\text{H}_2][\text{CO}] / [\text{H}_2\text{O}][\text{C}]$ | d. $K_{\text{eq}} = [\text{H}_2\text{O}] / [\text{H}_2][\text{CO}]$ |
- ___ 3. Which stress changes the value of K_{eq} ?
- | | |
|-------------------------|-----------------------------|
| a. Adding more reactant | b. Removing a product |
| c. Adding a catalyst | d. Changing the temperature |
- ___ 4. A precipitate will form when two solutions are mixed if
- | | |
|--|---|
| a. the ion concentration product is less than K_{sp} | b. the ion concentration product equals K_{sp} |
| c. the ion concentration product is greater than K_{sp} | d. the solutions have the same concentration |
- ___ 5. For the exothermic reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, which set of conditions would maximize NH_3 production?
- | | |
|--|--|
| a. High temperature, low pressure, low $[\text{N}_2]$ | b. Low temperature, high pressure, high $[\text{N}_2]$ |
| c. High temperature, high pressure, high $[\text{NH}_3]$ | d. Low temperature, low pressure, high $[\text{H}_2]$ |

True / False

Indicate whether the statement is true or false.

- ___ 6. At equilibrium, the concentrations of reactants and products are stable but not necessarily equal.
- ___ 7. Adding a catalyst to a system at equilibrium shifts the equilibrium to the right.
- ___ 8. For a reaction with equal moles of gas on both sides, changing pressure does not shift the equilibrium.
- ___ 9. $K_{\text{eq}} > 1$ means the forward reaction predominates; $K_{\text{eq}} < 1$ means the reverse reaction predominates.

- ___ 10. Increasing pressure on a gaseous system always increases the value of K_{eq} .
- ___ 11. The K_{sp} expression for a dissolving solid does NOT include the concentration of the solid itself.
- ___ 12. Removing a product from a system at equilibrium causes the equilibrium to shift to the left.
- ___ 13. For an endothermic reaction, increasing temperature shifts equilibrium to the right and increases K_{eq} .
- ___ 14. If the ion concentration product $< K_{sp}$, a precipitate will NOT form.
- ___ 15. The common-ion effect decreases the solubility of a salt in a solution already containing that ion.

Short Answer

16. Consider the reaction: $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta H = -198 \text{ kJ}$ a. Write the K_{eq} expression. b. Predict the shift direction when pressure is increased. c. Predict the shift direction and effect on K_{eq} when temperature is increased. d. Predict the shift direction when SO_3 is removed.

17. The K_{sp} of CaF_2 is 3.9×10^{-11} . $CaF_2 \rightleftharpoons Ca^{2+}(aq) + 2F^{-}(aq)$ Calculate the solubility of CaF_2 in mol/L. Show all work.

18. Define Le Châtelier's principle and give one real-world industrial example of how chemists use it to maximize the yield of a product.

Essay

Answer TWO of the following. You may answer all 3 for possible bonus.

19. Summarize all three sections of Chapter 17 in one comprehensive response. Your answer should cover: (a) what chemical equilibrium is and how K_{eq} is written, (b) Le Châtelier's principle and the three types of stresses, and (c) K_{sp} and how it relates to solution saturation.

20. A chemical engineer wants to maximize the production of SO_3 in the reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, which is exothermic. Using Le Châtelier's principle, describe what conditions of

temperature, pressure, and concentration the engineer should choose and explain the reasoning behind each choice.

21. Compare and contrast K_{eq} and K_{sp} . What do they have in common? How do they differ? How is each used to make predictions about a reaction? Give one worked example for each showing how the constant is used.